Syed Taha

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Education

Institute of Business Administration (IBA) Karachi, Pakistan
Bachelor of Science in Computer Science, GPA: 3.53/4.0 *Expected May 2027*

Experience

Teaching Assistant Aug 2024 - Present
Institute of Business Administration, Karachi *Karachi, Pakistan*

- Supported Introduction to Programming & OOP, Computer Architecture and Operating Systems by facilitating weekly C++ labs for 100+ students and giving one-on-one guidance on problem-solving and debugging.
- Supervised student projects, from 10 Computer Architecture teams building RISC-V assembly and hardware-optimized systems (S4 models for galaxy morphology classification) to Operating Systems projects extending the xv6 LLM runtime into distributed inference, covering OS security, distributed inference security, and networked operating systems.
- Graded assignments and assessments and acted as liaison between faculty and students, clarifying requirements and keeping academic standards consistent.

Data Science Intern Aug 2026 - Sep 2026
Foretheta, LLC *Seattle, WA (Remote)*

- Diagnosed and resolved data-deduplication bugs in a search reranking pipeline, improving result quality while preserving system guarantees around output completeness; added automated test coverage for the fix.
- Built and maintained a statistical evaluation framework for comparing search-ranking algorithm variants, using paired bootstrap confidence intervals to distinguish genuine improvements from statistical noise.
- Performed root-cause analysis of search ranking quality issues across a large evaluation dataset, tracing problems back to specific bugs in request construction and model input handling.
- Designed and ran a statistically rigorous comparison of alternative machine learning models for a domain-specific retrieval task, including a power analysis to size the evaluation appropriately and a licensing review of candidate models.

Software Engineering Fellow Jul 2024 - Sep 2024
Headstarter AI *Long Island City, NY (Remote)*

- Delivered 5 AI-powered web apps in 5 weeks using React/Next.js, Firebase, Clerk, and Vercel under weekly agile sprints with CI/CD, scaled each to 200+ users and launched an IBA teacher-review portal with 300 waitlist sign-ups.
- Co-developed a Next.js customer-support agent with a custom OpenAI+Pinecone RAG pipeline and a GPT-4o-powered flashcard SaaS, with weekly innovation sessions with engineers from Google, Y Combinator, Stanford, and Amazon.

Projects

Meridian: UCI Chess Engine | *C++*

- Wrote a UCI chess engine in C++20 that plays at about 2930 Elo on Stockfish's UCI_Elo scale (1 second per move, 250 games) and searches 4.2 million nodes per second on one core.
- Cut a depth-6 search from 1,609 ms to 0.93 ms (1,700x) with magic bitboards, move ordering, a transposition table, null-move and late-move pruning, keeping each technique only if benchmarks showed a gain.
- Trained a neural network evaluation on 38.7 million positions (16.9 million Leela Chess Zero, 21.8 million self-play) and verified the engine against published perft counts and 1,404 mate puzzles.

Othello AI with MCTS & Optimized CNNs | *C++, Python, PyTorch*

- Built an AlphaZero-style Othello engine: Monte Carlo tree search guided by a CNN trained through self-play, then distilled from 18.7 million to 4.8 million parameters (19.4 MB) to run on one CPU core.
- Rewrote the rules, network and search in C++ with AVX2 kernels: legal moves take 7.7 ns (1,600x to 2,400x faster than Python), a network evaluation 200 microseconds per board (13.5x faster than PyTorch), and a 500-simulation move about 100 ms.

OctaC: Statically Typed Language and Compiler | *C++, Python*

- Designing and implementing a compiler for a statically typed, C-structured language for numerical and matrix computation with built-in vector and matrix types.

- Completed the DFA-driven lexer and the parser with a pytest suite; semantic analysis and code generation follow, documented in a LaTeX technical book.

LLM Inference in the xv6 Educational OS | *C, RISC-V, xv6*

- Engineered a POSIX-compliant Shared Memory (SHM) subsystem for the xv6 kernel to enable zero-copy, persistent caching of Large Language Model weights (about 60MB), eliminating redundant network transfers and reducing subsequent inference latency from about 60s to sub-second.
- Designed a cycle-accurate performance profiling library that tracks call hierarchies and identified matrix multiplication as the dominant bottleneck (consuming 87-92% of inference time), providing a data-driven basis for optimization.
- Successfully ported the LLaMA-2 inference engine to xv6, resolving system-level challenges (stack overflow via iterative quicksort) and replacing Linux dependencies, achieving stable throughput of about 16 tokens/sec on RISC-V.

Pariah: GNN-Transformer Intrusion Detection | *Python, PyTorch*

- Co-authoring a systematization-of-knowledge study of whether GNN-Transformer hybrids stay reliable for intrusion detection when the deployment network differs from the training network, with a taxonomy of graph-based detectors.
- Testing invariance objectives (IRM, EERM, DIR) against tuned ERM and Group DRO under pre-stated hypotheses, using 10+ seeds per configuration, paired bootstrap confidence intervals, and Holm correction.

Project GLIDE: GPS-Denied Drone Navigation | *Python, PyTorch*

- Conducting research on GPS-free drone navigation by magnetic field: whether a drone carrying only a 3-axis magnetometer can navigate a region using the Earth's magnetic field, supported by our own ground surveys.
- Investigating a small S4D state space model as the pilot in a physics-informed simulator, in closed loop with its own predicted position and heading as the next state, so that physical laws govern navigation instead of a purely machine learning approach.

PiSSM: Distributed State Space Model Inference | *Python, PyTorch, gRPC, FastAPI*

- Built a distributed inference system on six Raspberry Pi 5 boards (24 GB total, $1,530) that runs Mamba-based large language models no single 4 GB board can hold.
- Split the model across boards with pipeline parallelism, at only 1.1 to 1.2x the latency of a single node for 30-token generation at prompt lengths of 64 to 1,024 tokens.
- Chose state space models over Transformers to keep communication at a flat 16 to 20% of total time as prompts grow, and added heartbeat-based failure detection.

Vectorized CNN Inference on 32-bit RISC-V Cores | *RISC-V Assembly*

- Implemented a complete Convolutional Neural Network in RISC-V assembly, leveraging the Vector Extension (RVV) to optimize core operations (convolution, pooling, softmax) for edge computing constraints.
- Achieved a 7.3x speedup in total inference time (8.223s to 1.122s) and a 12.2x reduction in convolution instructions via algorithmic redesign (im2col + GEMM) and FP16 compression.

Technical Skills

Languages: C++, C, Python, RISC-V Assembly, Bash, JavaScript
Frameworks & Tools: PyTorch, React, Next.js, FastAPI, gRPC, Google Cloud, Git, GitHub Actions
High-Performance Computing: CUDA (Basic), Vectorization, Multi-threading, Performance Profiling

Achievements

- Achieved 1st place in the *"Enhanced Safe Driver Prediction"* Kaggle competition among 100+ participants.
- Achieved 1st place in the *"Hello World"* competitive programming competition organized by CSS at IBA.